OFFLINE-FIRST SECURITY ENGINEERING

VNAGY Use Case — Financial Behavioral Anomaly Detection

Document ID: VNAGY-UCF-001

Version: v1.0

Revision History:

Version	Date	Author	Description
V1.0	19 August 2026	V. Nađ	Draft

Document Status: Internal Technical Specification

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1. Executive Summary

This document describes how the **VNAGY** Framework conceptually supports financial environments through behavior-driven analysis of processes, signals, and heuristics. **VNAGY** operates offline-first, without integration into core banking systems and without access to sensitive financial data. The goal is to reduce operational, security, and regulatory risk by detecting early anomalies in system behavior.

2. Business Context

Financial institutions rely on predictable, stable, and auditable system behavior. Behavioral anomalies often precede incidents such as fraud, unauthorized access, system compromise, or operational disruption.

3. Problem Statement

Typical behavioral risks in financial systems include:

- Unusual API usage patterns
- Memory-only processes interacting with critical modules
- Irregular timing profiles
- Network bursts to unknown destinations
- Abnormal CPU/memory behavior
- Unexpected process lineage

4. VNAGY Role

VNAGY:

- Analyzes process and signal behavior
- Correlates heuristics (CPU, network, timing, lineage, memory)
- Detects deviations from normal operational patterns
- Assigns deterministic risk levels
- Operates offline-first without integration

5. Workflow Overview

- A financial system generates an event.
- VNAGY collects the signal via the local event bus.
- Heuristics evaluate behavioral deviation.
- VNAGY assigns a risk level.
- The system receives a recommendation.

6. Expected Outcomes

- Early anomaly detection
- Reduced operational risk
- Improved auditability
- Increased transparency
- Enhanced stability

7. Risk Reduction

VNAGY reduces:

- Operational risk
- Security risk
- Regulatory risk
- Reputational risk

8. Conclusion

VNAGY provides stable, predictable, and secure behavior-driven analysis for financial environments without integration or access to sensitive data.

9. Licensing Reference

Subject to **VNAGY-PSDL-1.3** licensing constraints.

https://github.com/VNAGY-sec-ViktorijaN/VNAGY-Framework/blob/main/VNAGY/LICENSE/VNAGY%20Minimal%20License%20(PSDL%E2%80%911.3).pdf